

Task 2: Deep Learning – Image Classifier (CIFAR-10)

Internspark AI Internship

1. GitHub Repository

<https://github.com/zom1xo/cifar10-image-classifier>

2. Model File (Google Drive)

[<https://drive.google.com/file/d/1ZsA8iMEi4NEhdCJ1VkEeBg1WsxzNhCs/view?usp=sharing>]

3. Summary

Built a CIFAR-10 image classifier using transfer learning with pretrained ResNet18. Modified the architecture for 32×32 images. Applied data augmentation (RandomFlip, RandomRotation). Trained for 10 epochs on GPU. Achieved 91.77% test accuracy with strong generalization (train-val gap ~5.7%). Most common confusions: cat-dog (187), car-truck (50), bird-cat (46) — consistent with visual similarity between these classes.

4. Results

Metric	Value
Train Accuracy	96.79%
Val Accuracy	91.08%
Test Accuracy	91.77%
Train Loss	0.0937
Val Loss	0.2790

5. Key Visuals

- Training/validation loss and accuracy curves
- Confusion matrix (10×10)
- Sample predictions with correct/incorrect labels

6. Environment

- Python 3.x, Google Colab (GPU T4)
- PyTorch, torchvision, scikit-learn, matplotlib, seaborn
- Pretrained model: ResNet18 (ImageNet weights)